

A
Community Service Project Report

on
PLANTATION

Submitted in partial fulfilment of the requirements for the award of the degree of

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IN
ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

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ANNAMACHARYA INSTITUTE OF TECHNOLOGY AND SCIENCES

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CERTIFICATE

This is to certify that the community service project report titled “**PLANTATION**” submitted to “**ARTIFICIAL INTELLIGENCE AND DATA SCIENCE**” In **ANNAMACHARYA INSTITUTE OF TECHNOLOGY AND SCIENCES** is a bonfide record of work done under my supervision.

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We hereby declare that this Community Service Project report titled “**Plantation**” has been written by us. The work carried out is original and has not been submitted to any other University or Institutions for the award of any credit.

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Community Service Project Report

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TABLE OF CONTENTS

CHAPTER NO.	CHAPTER NAME	PAGE.NO
	List of Figures	i
	ABSTRACT	
CHAPTER 1	INTRODUCTION	01-02
CHAPTER 2	HISTROY OF PLANTATION	03-06
CHAPTER 3	ECOLOGICAL IMPACT ON PLANTATION	07-10
CHAPTER 4	CHARACTERISTICS OF PLANTATION	
4.1	Commercial Aspects	11
4.2	Large Scale Operation	11
4.3	Monocultures	11
4.4	Innovation and Development	12
4.5	Positives of Plantation Agriculture	12
4.6	Job Opportunities	12
4.7	Income to Governments	12
4.8	Modern Development	12-13
4.9	Colonialism	13
CHAPTER 5	PLANTATION SLAVE ECONOMY	
5.1	Plantation economy	14-15
5.2	Plantation Slave	15-16
CHAPTER 6	BENEFITS OF PLANTATION	
6.1	Purifying air	17
6.2	Cooling Down the Streets	17
6.3	Natural Air Conditioning	17
6.4	Saving Water	18

6.1	Purifying air	17
6.2	Cooling Down the Streets	17
6.3	Natural Air Conditioning	17
6.4	Saving Water	18
6.5	Preventing Water Pollution	18
6.6	Providing Shelters for Wildlife	18
6.7	Renewable Energy Source	18
6.8	Reinforcing Soil	19
6.9	Erosion Control	19
CHAPTER 7	CONCLUSION	20-21
CHAPTER 8	REFERENCE	22
CHAPTER 9	GALLERY	23-24

LIST OF FIGURES

S.NO.	FIGURE NAME	FIG.NO.	PAGE NO
1.	Planting a tree	1	2
2.	Plantation System	2	6
3.	Life Cycle Of A Forest	3.1	9
4.	Plantation Crop maintainence	3.2	10
5.	Planting a tree	3.3	10
6.	Plantation Economy	5.1	15
7.	Slaves Growing Sugar Trees	5.2	16

ABSTRACT

Large plantations producing tropical cash crops based on hired labor represent a sharp contrast with small family farms, popularly called “peasants” in developing economies. The family farm is an old institution that has existed since time immemorial, but the plantation is a new institution introduced by Western colonialism for extracting tropical cash crops for export to home country. Large-scale operation of the plantation was necessary for internalizing gains from investment in infrastructure needed for opening vast tracts of unused lands. However, where the communities of indigenous smallholders had already been established, family farms proved to be equally or more efficient producers of tropical export crops using the family labor of low supervision costs, relative to plantations based on hired labor. This advantage of family farms rose as population density increased and rural infrastructure improved, whereas not only economic but also social drawbacks of the plantation system loomed. However, reforms aimed to break down plantations to the operation of smallholders by a government's coercive power could be disruptive and inefficient. A better approach might be to support the initiative of the private sector to reorganize the plantation system into a more decentralized system, such as the contract farming system in which an agribusiness enterprise manages the processing/marketing process and contracts with small growers on the assured supply of farm-produced raw materials.

CHAPTER-1

INTRODUCTION

1. INTRODUCTION

Tree plantation is the process of transferring of seedlings from one place to another for different purposes. Furthermore, there are various reasons behind tree plantation but the most important are forestry, land reclamation, and landscaping. Each process of tree plantation is important and unique in its own way.

The most common and known purpose of tree plantation is forestry. The forest is very important for maintaining the ecological balance of the environment of the earth. Also, earlier the forest use to cover a major part of the surface of the earth. But, now due to the rapid cut down of forest due to industries, and land for settlement the number of forests has reduced. A part from that, the forest grows naturally but in order to increase the speed of reforestation, our contribution is a must for it. Also, it will help the forest to recover from the loss due to deforestation. Furthermore, with our help, the pace of tree plantation can be increased.

Plantation in the tropics, employing a large number of laborers under the Plantations command of centralized management hierarchy for the production of tropical cash crops for commercial sales, represent a unique organization in agriculture. They are unique in contrast with small farms based mainly on family labor, which are the common form of agricultural production in the world, including both temperate and tropical areas, with respect to subsistence crops such as rice, wheat, and maize. How has this organization emerged in the tropics? What roles has it played until present and might it play in the future? These are the questions addressed in this chapter.

According to a broad definition by William Jones (1968, p. 154), the plantation is “an economic unit producing agricultural commodities (field crops or horticultural products, but not livestock) for sale and employing a relatively large number off unskilled laborers whose activities are closely supervised.” This definition can include estate farms based on forced labor, such as slavery, corvee, and serf, instead of free wage labor and may also be called plantations. These estates based on forced labor were established typically before the onset of the industrial revolution in Europe and North America, having major impacts on the southern United States and Latin America, including the Caribbean as well as Eastern Europe.

Despite their great historical interests, these estate farms based on forced labor are not included in this review. The plantations dealt with in this chapter are those based on free wage labor, which were typically established in tropical Asia and Africa after the mid-19th century. From the beginning their operation was based on wage laborers, though many laborers were imported from densely populated economies such as China and India under long-term contracts, often tied by credit, akin to debt peonage.

Following this introduction, Section 2 outlines advantages of family farms under the specific environmental and technical conditions specific to agricultural production, to understand why plantations remain rather an exceptional organization in agriculture relative to family farms. Section 3 explains the unique advantage of the plantation system at the stage of opening new lands for cultivation of crops corresponding to the sudden expansion of external demand. Section 4 specifies the tendency of the plantation system to lose its advantage in the process of closing land frontiers, increasing population density, and improving rural infrastructure. Finally, Section 5 tries to identify the direction of reorganizing plantations to better serve the purpose of rural development in tropical developing economies.



Fig.1: planting a tree

CHAPTER-2

HISTORY OF PLANTATION

2. HISTORY OF PLANTATION

The term plantation arose as settlements in the southern United States, originally linked with colonial expansion, came to revolve around the production of agriculture. The word plantation first appeared in English in the 15th century. Originally, the word “plantation” was a verb that meant “to plant.” However, what came to be known as plantations became the center of large-scale enslaved labor operations in the Western Hemisphere. Historians Peter H. Wood and Edward Baptist advocate for calling these places “labor camps” or “slave labor camps” rather than “plantations,” in order to avoid hiding the truth of what really happened in these agricultural operations using forced labor.

The plantation system developed in the American South as British colonists arrived in what became known as Virginia and divided the land into large areas suitable for farming. The land on which these plantations were established was stolen from Indigenous nations through canceled, disregarded, and deceitful treaties, or outright violence.

The plantation system came to dominate the culture of the South, and it was rife with inequity from the time it was established. In 1606, King James I formed the Virginia Company of London to establish colonies in North America, but when the British arrived, they faced a harsh and foreboding wilderness, and their lives became little more than a struggle for survival. So, to make settling the land more attractive, the Virginia Company offered any adult man with the means to travel to America 50 acres of land. At the encouragement of the Company, many of the settlers banded together and created large settlements, called hundreds, as they were intended to support 100 individuals, usually men who led a household.

The hundreds were run as private plantations intent on making a profit from the cultivation of crops, which the economy of the South depended on. The plantation system was an early capitalist venture. Unlike small subsistence farms, plantations were created to grow cash crops for sale on the market. Tobacco and cotton proved to be exceptionally profitable. Cheap labor was used to cut production costs and maximize profits. Initially, the land was worked by indentured servants, who were mostly from England (and sometimes from Africa), and enslaved African and (less often) Indigenous people.

Indentured servants were contracted to work four- to seven-year terms without pay for passage to the colony, room, and board. After completing the term, they were often given land, clothes, and provisions.

The plantation system created a society sharply divided along class lines. In the colonies south of Pennsylvania and east of the Delaware River, a few wealthy, white land owners owned the bulk of the land, while the majority of the population was made up of poor farmers, indentured servants, and the enslaved.

Plantation labor shifted away from indentured servitude and more toward slavery by the late 1600s, in part because obtaining indentured servants became more difficult as more economic opportunities became available to them. Wealthy landowners also made purchasing land more difficult for former indentured servants. This sharpened class divisions, as a small number of people owned larger and larger plantations. Wealthy landowners got wealthier, and the use of slave labor increased. This led to uprisings and skirmishes with impoverished Black and white people joining forces against the wealthy.

In response, customs changed and laws were passed to elevate the status of poor white people above all Black people. This new class acted as a buffer to protect the wealthy, and Black people in the British American colonies were further oppressed. Thus, people of African descent were forced into a permanent underclass.

Despite this brutal history, plantations are not always seen as the violent places they were. For some, the word plantation suggests an idyllic past. This is seen at some of the United States plantations themselves, with some tours and tourists focusing on the wealth and lives of the enslavers while ignoring those they enslaved, and plantations continuing to be rented out as venues for weddings and parties. However, nationwide reckoning with the legacy of slavery has caused a cultural shift, spurring some plantation tours to present a more accurate view of the horrors that took place at these sites.

The romanticized notions of plantation life largely stem from an ideology called the Lost Cause which became popular shortly after the United States Civil War. The Confederates seceded from the United States to maintain the system of slavery.

After losing the war, many Confederates and Confederate sympathizers promoted an ideology that falsely altered the reason for secession. This ideology was named the Lost Cause after an 1866 book by Edward A. Pollard, a newspaper editor from Virginia who supported the Confederacy.

According to Caroline Janney, professor of History of the American Civil War at the University of Virginia, the Lost Cause ideology puts the Confederates in a favorable light.

She says the Lost Cause claims:

- 1) Confederates were patriots fighting to protect their constitutionally granted states' rights.
- 2) Confederates were not fighting to protect slavery.
- 3) Slavery was a benevolent institution in which Black people were treated well.
- 4) Enslaved Black people were faithful to their enslavers and happy to be held in bondage.
- 5) Confederate General Robert E. Lee and, to a lesser extent, General Thomas "Stonewall" Jackson were godlike figures. None of these claims are true. The Lost Cause perpetuates harmful and false narratives.

Besides Pollard's book, other works have carried the Lost Cause lie, including the 1864 painting, the "Burial of Latané" by William Washington, Thomas Dixon Jr.'s 1905 novel and play *The Clansman*, and Margaret Mitchell's 1936 novel *Gone with the Wind*. The last two became popular movies; *The Clansman* became *The Birth of a Nation*. Lost Cause propaganda was also continued by former Confederate General Jubal Early as well as various organizations of upper- and middle-class white Southern women—the Ladies Memorial Associations, the United Confederate Veterans, and the United Daughters of the Confederacy.

The United States wasn't the only participant in the transatlantic slave trade. Throughout the Western Hemisphere, the plantation served as an institution in itself, characterized by social and political inequality, racial conflict, and domination by plantation owners, known as the planter class. In the Caribbean, as well as in the slave states, the shift from small-scale farming to industrial agriculture transformed the culture of these societies, as their economic prosperity depended on the plantation.

Until the transatlantic slave trade was abolished in 1807, over 12 million Africans were transported to the “New World” “90 percent of them were sent to the Caribbean and South America to work on sugar plantations.



Fig.2: Plantation System

CHAPTER-3
ECOLOGICAL IMPACT
ON PLANTATION

3. ECOLOGICAL IMPACT ON PLANTATION

Probably the most critical factor a plantation has on the local environment is the site where the plantation is established. In Brazil, coffee plantations would use slash-and-burn agriculture, tearing down rainforests and planting coffee trees that depleted the nutrients in soil. Once the soil had been sapped, growers would move on to another place. If a natural forest is cleared for a planted forest, then a reduction in biodiversity and loss of habitat will likely result. In some cases, their establishment may involve draining wetlands to replace mixed hardwoods that formerly predominated with pine species. If a plantation is established on abandoned agricultural land or highly degraded land, it can increase both habitat and biodiversity. A planted forest can be profitably established on lands that will not support agriculture or suffer from a lack of natural regeneration.

The tree species used in a plantation are also an important factor. Where non-native varieties or species are grown, few native faunas are adapted to exploit these, and further biodiversity loss occurs. However, even non-native tree species may serve as corridors for wildlife and act as a buffer for native forests, reducing edge effect.

Once a plantation is established, managing it becomes an important environmental factor. The most critical aspect of management is the rotation period. Plantations harvested on more extended rotation periods (30 years or more) can provide similar benefits to a naturally regenerated forest managed for wood production on a similar rotation. This is especially true if native species are used. In the case of exotic species, the habitat can be improved significantly if the impact is mitigated by measures such as leaving blocks of native species in the plantation or retaining corridors of natural forest. In Brazil, similar measures are required by government regulation.

Plant ecology is a subdiscipline of ecology that studies the distribution and abundance of plants, the effects of environmental factors upon the abundance of plants, and the interactions among plants and between plants and other organisms. Examples of these are the distribution of temperate deciduous forests in North America, the effects of drought or flooding upon plant survival, and competition among desert plants for water, or effects of herds of grazing animals upon the composition of grasslands.

A global overview of the Earth's major vegetation types is provided by O.W. Archibold. He recognizes 11 major vegetation types: tropical forests, tropical savannas, arid regions (deserts), Mediterranean ecosystems, temperate forest ecosystems, temperate grasslands, coniferous forests, tundra (both polar and high mountain), terrestrial wetlands, freshwater ecosystems and coastal/marine systems. This breadth of topics shows the complexity of plant ecology, since it includes plants from floating single-celled algae up to large canopy forming trees.

One feature that defines plants is photosynthesis. Photosynthesis is the process of a chemical reactions to create glucose and oxygen, which is vital for plant life. One of the most important aspects of plant ecology is the role plants have played in creating the oxygenated atmosphere of earth, an event that occurred some 2 billion years ago. It can be dated by the deposition of banded iron formations, distinctive sedimentary rocks with large amounts of iron oxide. At the same time, plants began removing carbon dioxide from the atmosphere, thereby initiating the process of controlling Earth's climate. A long-term trend of the Earth has been toward increasing oxygen and decreasing carbon dioxide, and many other events in the Earth's history, like the first movement of life onto land, are likely tied to this sequence of events.

One of the early classic books on plant ecology was written by J.E. Weaver and F.E. Clements. It talks broadly about plant communities, and particularly the importance of forces like competition and processes like succession. The term ecology itself was coined by German biologist Ernst Haeckel.

Plant ecology can also be divided by levels of organization including plant ecophysiology, plant population ecology, community ecology, ecosystem ecology, landscape ecology and biosphere ecology.

First, most plants are rooted in the soil, which makes it difficult to observe and measure nutrient uptake and species interactions. Second, plants often reproduce vegetatively, that is asexually, in a way that makes it difficult to distinguish individual plants. Indeed, the very concept of an individual is doubtful, since even a tree may be regarded as a large collection of linked meristems.

Hence, plant ecology and animal ecology have different styles of approach to problems that involve processes like reproduction, dispersal and mutualism. Some plant ecologists have placed considerable emphasis upon trying to treat plant populations as if they were animal populations, focusing on population ecology. Many other ecologists believe that while it is useful to draw upon population ecology to solve certain scientific problems, plants demand that ecologists work with multiple perspectives, appropriate to the problem, the scale and the situation.

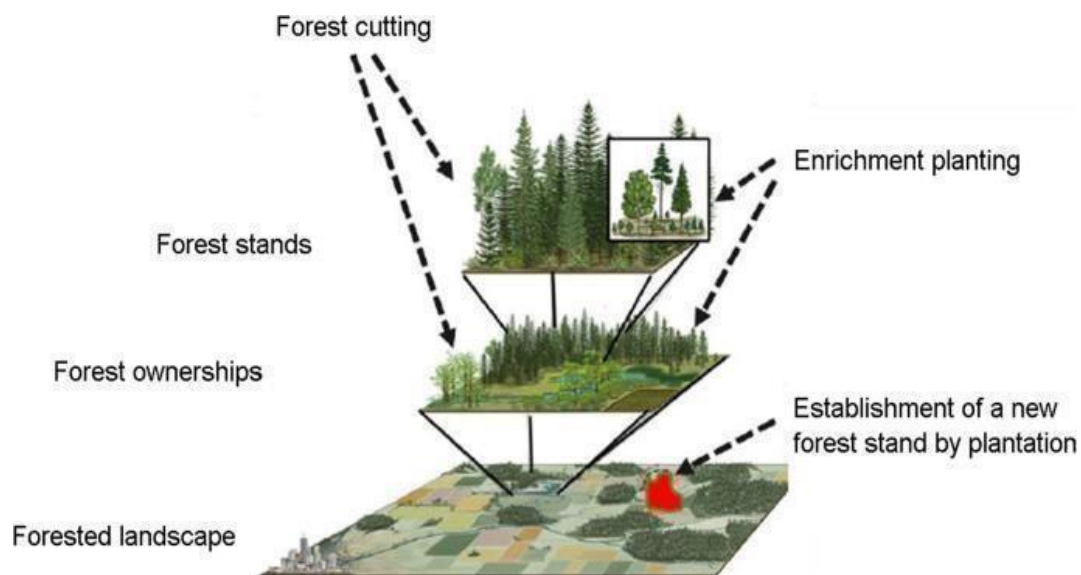


Fig.3.1: Life Cycle of a Forest



Fig.3.2: Plantation Crop Maintenance



Fig.3.3: Plantinng a Tree

CHAPTER-4
CHARACTERISTICS OF
PLANTATION

4. CHARACTERISTICS OF PLANTATION

There is a large range of characteristics associated with plantation agriculture. Let's take a look at some of these characteristics.

4.1 Commercial Aspects:

Plantations are very much commercial in the sense that the products grown on plantations are cash crops. These crops are grown in mostly low and middle-income countries as exports to gain foreign investment, typically from countries in North America and Europe. High yields, therefore, generate high amounts of money, which is key for the commercial aspect of plantations.

4.2 Large-Scale Operation:

Plantations occur on a huge scale and require large amounts of work in order to produce high-quality crops regularly to meet high demands. Such a commercial process means large yields of the crops are grown, requiring numerous employees. These employees are laborers, who work on the plantation working long hours, mostly harvesting the crop.

4.3 Monocultures:

Plantations are essentially monocultures are when one crop is grown in one area of agricultural land .Monocultures are necessary aspects of plantations because it allows increased efficiency of planting, harvesting, and processing, as only one type of crop is grown across the entire plantation.

However, monocultures can create environmental issues as they can cause the spreading of diseases and pests since there is only one type of crop grown. This ultimately reduces soil quality and leads to a loss of biodiversity. This can have negative impacts on crop growth and therefore turnover of yield, leading to a loss in profit for plantation owners.

4.4 Innovation and Development:

Plantations are aided by well-developed transportation and strong communication networks. This combined with the economic gain of plantations leads to the research and development of machinery that is used in plantations to enhance the processing and speed of crop growth and harvesting. Many plantations use this advanced machinery, which allows for fast turnover of crop and therefore a large economic gain.

4.5 Positives of Plantation Agriculture:

Plantation agriculture is deemed important due to a variety of factors. These include; job opportunities, income to governments, and modern development.

4.6 Job Opportunities:

Plantation agriculture provides many job opportunities and income to locals. Plantations are most commonly located in developing countries; therefore, it may be hard for many citizens to find work and receive an income. This is due to challenges such as poor working conditions, low wages, wage gaps, and workplace discrimination. However, plantations provide employment opportunities for locals, including labor work such as growing, harvesting, and processing the crop. This ensures a steady income for the workers.

4.7 Income to Governments:

Plantation agriculture also offers income to the government as it is a source of foreign trade. This is because external companies from foreign countries may use the land as plantations and export the crops, which provides income to the country via foreign revenue. This is vital for developing countries, helping them to become more connected due to globalization and economic gain.

4.8 Modern Development:

Plantations increase modern development and industrial growth. Since plantations occur on a mass agricultural scale, the development of technology and machinery to increase processing times is needed.

This encourages the growth of based processing industries. Based industries are industries that produce raw agricultural materials. Plantations also encourage further agricultural development and research, such as growing strains of crops that are resistant to diseases and pests. Disease-resistant crops are vital in plantation agriculture because if one crop develops a disease, all the surrounding crops also develop the disease due to the proximity of the fields and because they are the same type of crop. Therefore, developing a strain of crop resistant to the disease allows all the crops to grow healthy.

4.9 Colonialism:

The history of plantations is associated with colonialism. This is because plantations were set up by British colonialists during the Colonial Era (between the 15th and 19th centuries). Large areas of land deemed suitable for farming were turned into plantations, and the exploitation of slave labor occurred. It is considered that plantations are still exploitative due to the fact companies utilize.

CHAPTER-5
PLANTATION SLAVE
ECONOMY

5.1 PLANTATION ECONOMY

In Plantation owners extensively used enslaved Africans to work on early plantations (such as tobacco, rice, cotton, hemp, and sugar plantations) in the American colonies and the United States, throughout the Caribbean, the Americas, and in European- occupied areas of Africa.

modern times, the low wages typically paid to plantation workers are the basis of plantation profitability in some areas.

In more recent times, overt slavery has been replaced by para-slavery or slavery-in-kind, including the share cropping system, and even that has been severely reduced. At its most extreme, workers are in “debt bandage”: they must work to pay off a debt at such punitive interest rates that it may never be paid off. Others work unreasonably long hours and are paid subsistence wages that (in practice) may only be spent in the company store.

In Brazil, a sugarcane plantation was termed an *engenho* ("engine"), and the 17th-century English usage for organized colonial production was "factory." Such colonial social and economic structures are discussed at plantation economy.

Sugar workers on plantations in Cuba and elsewhere in the Caribbean lived company towns known as *bat eyes*.



Fig.5.1: Plantation Economy

5.2 PLANTATION SLAVE:

Planters embraced the use of slaves mainly because indenture labor became expensive. Some indentured slaves were also leaving to start their farms as land was widely available. Colonists in the Americas tried using Native for labor, but they were susceptible to European diseases and died in large numbers. The plantation owners then turned to enslaved Africans for labor. In 1665, there were fewer than 500 Africans but by 1750, 85 percent of the 235,000 slaves in the Thirteen colonies lived in the southern colonies, Virginia included. Africans made up 40 percent of the South's population.

According to 180 united states cencius,one out of every four families in Virginia owned slaves. There were over 100 plantation-owners who owned over 100 slaves. The number-øf slaves in the 15 States was just shy of 4 million in a total population of 12.4 million and the percentage was 32% of the population.

- Number of slaves in the Lower South: 2,312,352 (47% of total population) 4,919 million.
- Number of slaves in the Upper South: 1,208,758 (29% of total population) 4,165 million.
- Number of slaves in the Border States: 432,586 (13% of total population) 3,323 million.

Fewer than one-third of Southern U.S. families owned slaves at the peak of slavery prior to the outbreak of the American civil war in 1861. In the figure approached one-half. The total number of slave owners was 385,000 some accounting to approximately 3.8% of the Southern and population.



Fig.5.2: Slaves Growing Sugar Trees

CHAPTER-6

BENEFITS OF PLANTATION

6. BENEFITS OF PLANTATION

If people are good at something, then it is building up excess carbon dioxide in the atmosphere. Harmful CO₂ contributes to climate change, the biggest current problem the world has to deal with. Trees, however, help fight it. They absorb CO₂ removing it from the air and storing it while releasing oxygen. Annually, an acre of trees absorbs the amount of carbon dioxide equal to driving your car 26 000 miles. Trees are our main survival tools; only one tree can produce enough oxygen for four people.

6.1 Purifying Air:

Every year we listen to the shocking global warming news. For instance, the average temperature in Los Angeles has risen by 6F in 50 years, and the average global temperature grew by 1.4 F. This happens as tree coverage declines. Removing trees and replacing them with heat absorbing asphalt roads and buildings makes cities much warmer. Trees are cooling cities by up to 10 F by providing shade and releasing water.

6.2 Cooling Down the Streets:

Every year we listen to the shocking global warming news. For instance, the average temperature in Los Angeles has risen by 6F in 50 years, and the average global temperature grew by 1.4 F. This happens as tree coverage declines. Removing trees and replacing them with heat absorbing asphalt roads and buildings makes cities much warmer. Trees are cooling cities by up to 10 F by providing shade and releasing water.

6.3 Natural Air Conditioning:

Did you know that strategically placed trees around your home can significantly cut air conditioning needs? Not only will this make your wallet thicker, but it will also reduce carbon dioxide and emissions from power plants. Architects and environmentalists sat together and came up with the great solution – green roofs. Green roofs are an amazing way to incorporate vegetation to your home and provide environmental benefits for your community while saving money on cooling bills.

6.4 Saving Water:

Except for cooling, trees also help to save water. Because of the shade they provide, water will evaporate slowly from low vegetation. Trees need about 15 water gallons a week to survive, and they release about 200-450 gallons of water per day.

6.5 Preventing Water Pollution:

Stormwater can be full of phosphorus pollutants and nitrogen. Without trees, stormwater flows into oceans and waters without being filtered. Trees break the rainfall and allow water to enter the earth and seep into the soil. Therefore, they prevent stormwater from polluting oceans. Except for trees, green infrastructure like green roof can help sooth effects of stormwater.

6.6 Providing Shelters for Wildlife:

Trees also contribute to boosting biodiversity as they become a food source and natural habitat for wildlife. One apple tree produces about 20 fruit bushels per year which can nourish many birds, insects, and wildlife. It can be planted on a very small surface but has a fantastic environmental effect. Trees that are most planted as homes for birds, squirrels, and are oak and sycamore.

6.7 Renewable Energy Source:

Nowadays, one of the biggest world problems is fossil fuels. Except for their toxic properties, they will also not be around forever. So many companies are focusing on renewable sources of energy. If harvested and treated sustainably, trees can become a great renewable source of energy. They are simple to use, have been around since the beginning of time, and with smart forest management they can become an excellent eco- friendly fuel.

6.8 Reinforcing Soil:

Trees are one of the best partners when it comes to agriculture. They act positively in several ways: they reduce soil erosion, increase fertility and help soil obtain moisture. Fallen tree leaves lower reduce soil temperature and prevent soil from losing too much moisture. Decaying leaves that fall onto the ground turn into nutrients for tree growth and promote microorganism development.

6.9 Erosion Control:

Rain and wind are two primary erosion forces that damage the bare soil. As they fall from heights, drops of rain gain power and momentum which is strong enough to penetrate soil once they hit the ground. On the other hand, if the land is dried out, then wind can do significant damage. Trees break droplets of rain and weaken their strength while roots hold the soil together and protect it from effects.

CHAPTER-7

CONCLUSION

7. CONCLUSION

The central conclusion from the analysis is that role of plantation forests, in meeting future wood and fiber demands, will increase during the next 30 years, irrespective of future rates of plantation establishment. Trees in the ground already largely determine wood supplies from forest plantation for the next decade. In many countries, a considerable increase in areas of plantation forests reaching harvestable age will occur. By 2010 the annual yield of plantation-grown industrial roundwood is estimated to increase from its current 414 million to around 600 million m³yr⁻¹.

Beyond 2010, plantation production forecasts are increasingly dependent on assumptions of new planting rates, and on assumptions of improvements in annual increments. There is scope, depending on policy decisions and markets, for forest plantations to play a dominant role in industrial wood and fiber supplies. A more likely scenario is that the proportion of forest plantation-grown wood will increase, but natural forests will continue to supply a modest majority of industrial roundwood.

The question of where future plantation forest development is most likely to occur remains unclear. At present, many governments remain active in plantation forest establishment, either directly, through state planting program, or indirectly, by providing incentives to the private sector. In some instance plantations realizing non-market values can justify incentives. In other case the incentives are merely maintaining wood supply capacity. In any event, under these circumstances, competitive and comparative advantages are not clearly emerging.

The most significant forest plantation increases in the immediate future will be in countries where specific public planting program are in force, most notably China and India. In Europe, plantation establishment is likely to be mainly dictated by the life span of EU incentive policies. Europe is largely self-sufficient in terms of wood-fiber volumes and development of a larger than present plantation-based export trade seems a unlikely development.

South America and Oceania are likely to continue to expand forest plantation areas under the perception that real competitive advantage in plantation growing is held in these regions. The extent to which forest plantations increase in these regions is likely to be dictated by the extent to which this perception is sustained. If plantation revenues are unsatisfactory then afforestation rates in these regions are likely to slow.

Plantation profitability in the southern countries is likely to be determined by conditions in North American and Asian markets and by wood and fiber supplies from forests in these regions. Similarly, if the current trend toward increasing regulation of natural forests increases then forest plantation establishment in both North America and Asia is also likely to accelerate. Of key interest will be natural forest wood supply trends in the US Pacific- Northwest, Canada, Indonesia and Malaysia.

Plantation forest establishment in the former-USSR and Africa seems unlikely to accelerate in the immediate future. In the former-USSR countries, economic difficulties are likely to mean plantation investment will be of relatively low priority, particularly given the extensive natural forest resource in several of these countries. In Africa, the absence of strong infrastructure is likely to remain a significant competitive disadvantage for many countries. It is difficult to see true competitive advantage in plantation investment emerging in many countries, even those that have important industries based on natural forests.

CHAPTER-8

REFERENCE

8. REFERENCES

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CHAPTER-9

GALLERY

9. GALLERY







Video link

https://drive.google.com/file/d/16NGQTWvFyXyDb89maq_sYe-a5EEJ1vfW/view?usp=drivesdk

